Data Analysis and Machine Learning Approaches for Predicting Movie Ratings

Final Report

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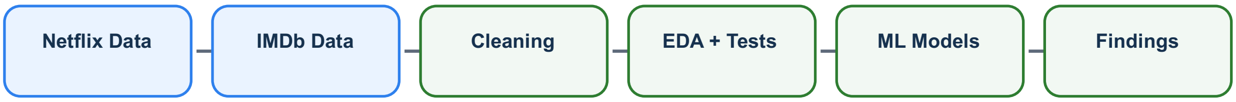
Project Summary

This project investigates which movie-related variables are associated with IMDb ratings by combining Netflix content information with IMDb audience feedback. The analysis follows a complete data science pipeline: data collection, cleaning, exploratory data analysis, hypothesis testing, machine learning model comparison, and interpretation of results.

Key result: Random Forest achieved the best predictive performance among the tested models, and number of voted users was identified as the most influential feature.



1. Motivation

Movie ratings are important indicators of audience satisfaction and movie success. Platforms such as Netflix and IMDb contain large amounts of information about movies, including genres, release years, durations, audience votes, and ratings. Analyzing these variables can help identify which factors influence movie ratings the most.

The motivation of this project is to explore relationships between movie characteristics and IMDb scores using data analysis and machine learning methods. In addition to exploratory data analysis and hypothesis testing, several machine learning models were applied and compared to predict movie ratings based on movie features.

This project aims to understand audience behavior, discover meaningful patterns in movie datasets, and evaluate the performance of different machine learning algorithms.

2. Data Source

The project uses two public datasets:

- Netflix movie dataset: movie titles, genres, release years, and durations.

- IMDb dataset: IMDb scores and audience vote counts.

The datasets were merged using movie titles after preprocessing steps such as lowercase conversion, title cleaning, and removal of missing values. This enrichment step allowed the project to combine content-related movie characteristics with audience feedback.

Variable	Description
imdb_score	Target variable; IMDb movie rating.
num_voted_users	Audience engagement measured by number of votes.
release_year	Movie release year.
duration_min	Movie duration in minutes.
primary_genre	Main genre extracted from Netflix categories.

Table 1. Main variables used in the analysis.

3. Data Preparation

- Only movie entries were selected from the Netflix dataset.
- Titles were converted to lowercase to improve matching between datasets.
- Missing values were removed after merging.
- Movie durations were converted into numeric minute values.
- The first listed genre was selected as the primary genre.
- Categorical genre values were transformed using one-hot encoding for machine learning models.

These preprocessing steps improved dataset consistency and made the data suitable for both statistical analysis and machine learning.

4. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was used to understand the structure of the dataset and identify relationships between movie characteristics and IMDb ratings.

- IMDb score distribution: most movies had medium to high scores, while very low scores were less common.
- Genre analysis: average IMDb scores differed across movie genres, suggesting that genre may influenceaudience ratings.
- Vote count and rating: movies with more audience votes generally tended to have higher ratings.
- Release year and duration: older movies showed slightly higher average ratings in this dataset, whileduration had a weaker relationship with rating.
- Correlation heatmap: vote count had the strongest relationship with IMDb score among the numericalfeatures.

Overall, the EDA suggested that both audience-related variables and content-related movie characteristics are relevant to movie ratings.

5. Hypothesis Testing

Hypothesis	Null Hypothesis (H0)	Result
Votes and IMDb score	No relationship between vote count and score	Rejected H0: significant positive relations
Genre and IMDb score	All genres have the same average score.	Rejected H0: genre affects ratings.
Old vs new movies	Older and newer movies have the same average score	Rejected H0: older movies had slightly higher scores.

Table 2. Summary of hypothesis testing results.

6. Machine Learning Analysis

Several machine learning models were applied to predict IMDb scores using movie-related features. The target variable was imdb_score, while the features included num_voted_users, release_year, duration_min, and one-hot encoded primary_genre variables.

The following models were compared:

- Linear Regression
- Ridge Regression
- Random Forest Regressor
- Gradient Boosting Regressor

The models were evaluated using RMSE, MAE, and R2 score. RMSE and MAE measure prediction error, so lower values indicate better performance. R2 measures how much of the variation in IMDb scores is explained by the model, so higher values indicate better performance.

Model	RMSE	MAE	R2
Linear Regression	0.7887	0.6105	0.3048
Ridge Regression	0.7886	0.6103	0.3050
Random Forest	**0.7553**	**0.5706**	**0.3624**
Gradient Boosting	0.7797	0.5756	0.3205

Table 3. Machine learning model comparison. Random Forest is highlighted as the best model.

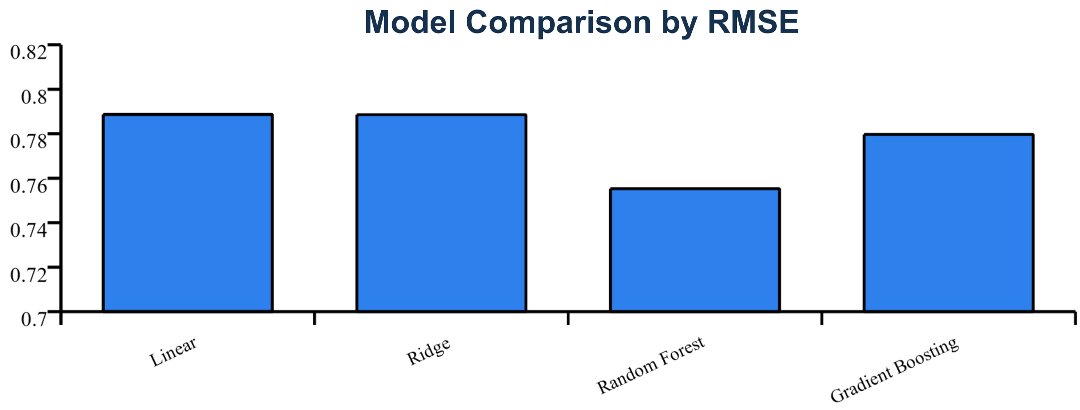


Figure 1. Model comparison by RMSE. Lower RMSE indicates better predictive performance.

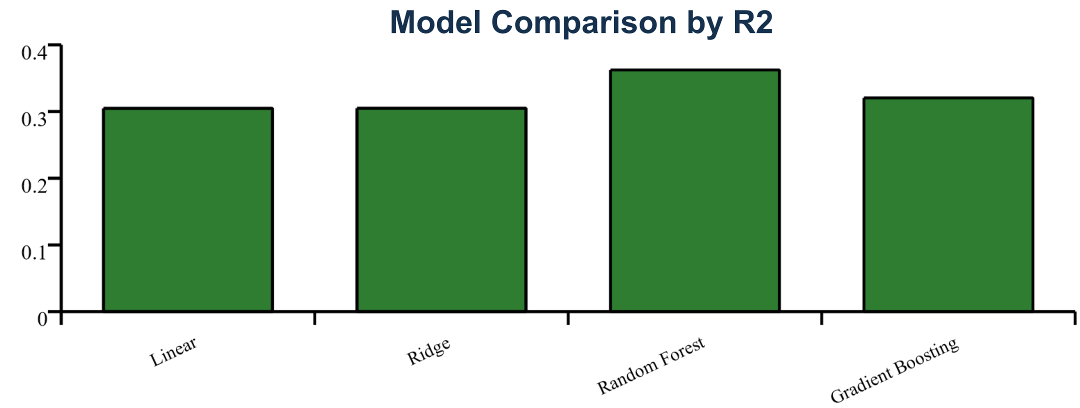


Figure 2. Model comparison by R2 score. Higher R2 indicates better explanatory power.

Random Forest had the lowest RMSE and MAE and the highest R2 score. This suggests that nonlinear ensemble methods captured relationships in the data better than simple linear models.

However, the R2 values are still moderate. This means the available features explain only part of the variation in IMDb scores. Ratings are also affected by variables not included in the dataset, such as actors, directors, marketing, cultural context, budget, and personal preferences.

7. Feature Importance

Feature importance analysis was conducted using the Random Forest model to understand which features contributed most strongly to prediction. The most important feature was the number of voted users, showing that audience engagement is strongly related to IMDb ratings.

Rank	Feature	Interpretation
1	num_voted_users	Audience engagement is the strongest predictor.
2	duration_min	Movie length contributes to rating differences.
3	release_year	Time period has some predictive value.
4	primary_genre_Dramas	Genre type adds smaller but useful information.

Table 4. Ranked feature importance interpretation based on the Random Forest model.

This result supports the earlier EDA and hypothesis testing findings, especially the relationship between vote count and IMDb score.

8. Findings

- Movies with more votes tend to have higher IMDb scores.
- Genre has a statistically significant effect on movie ratings.
- Older movies have slightly higher ratings in this merged dataset.
- Duration has a weaker but visible relationship with IMDb score.
- Random Forest performed better than linear models for rating prediction.
- Vote count was the most important feature in the machine learning analysis.

Overall, both audience-related variables and movie characteristics influence IMDb ratings.

9. Limitations and Future Work

This project has several limitations. First, the merged dataset became smaller after matching Netflix and IMDb movie titles. Some movies could not be matched because of title formatting differences or missing values. Second, the dataset includes limited features. Movie ratings may also depend on actors, directors, budget, language, country, marketing, and cultural factors. Third, IMDb ratings are based on user votes, so they may contain audience bias.

For future work, the dataset could be improved by adding director information, cast popularity, movie budget, country, language, and box office revenue. Cross-validation and hyperparameter tuning could also provide a more reliable evaluation of model performance.

10. Conclusion

This project analyzed movie ratings using Netflix and IMDb datasets. The analysis included data collection, cleaning, exploratory data analysis, hypothesis testing, and machine learning. The results showed that vote count, genre, release year, and duration are related to IMDb scores.

Among the machine learning models, Random Forest achieved the best performance. Feature importance analysis showed that number of voted users was the strongest predictor of IMDb score. In conclusion, movie ratings are influenced by both audience engagement and movie characteristics. Although the models provided useful insights, more detailed data would be needed to make stronger predictions.

11. AI Assistance Disclosure

AI tools were used during this project for guidance in code debugging, machine learning model development, visualization suggestions, and report structuring. All analysis, implementation decisions, and interpretations were reviewed and validated by the author.